WORK SHEET - 32 **SEQUENCE AND PROGRESSION**

Q.1 (B)

$S_n = n(n^2 - 1)$

$T_n = S_n - S_{n-1} = n(n-1)(n+1) - (n-1)(n-2)(n) = n(n-1)(n+1-n+2) = 3n(n-1)$

$$\therefore \sum_{n=2}^{\infty}\frac{1}{T_n} = \frac{1}{3}\sum_{n=2}^{\infty}\frac{1}{n(n-1)} = \frac{1}{3}\sum_{n=2}^{\infty}\left(\frac{1}{n-1}-\frac{1}{n}\right) = \frac{1}{3}\left(\frac{1}{1}-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\ldots\ldots\right) = \frac{1}{3} \quad \textbf{Ans.}$$

Q.2 (B)

$a_n - a_{n-1} = n^2$

Putting $n = 2, 3, 4, \ldots\ldots k$

$a_2 - a_1 = 2^2$

$a_3 - a_2 = 3^2$

$\vdots$

$a_k - a_{k-1} = k^2$

$a_k - a_1 = 2^2 + 3^2 + \ldots\ldots + k^2$

$a_k = 2 + 2^2 + 3^2 + \ldots\ldots + k^2 = 205$

$\Rightarrow \quad 1 + (1^2 + 2^2 + \ldots\ldots + k^2) = 205$

$$\Rightarrow \quad 1 + \frac{k(k+1)(2k+1)}{6} = 205$$

$k(k+1)(2k+1) = 6 \times 204 = 6 \times 3 \times 4 \times 17 = 8 \times 9 \times 17$

$\therefore \quad k = 8$ **Ans.**

Q. 3 (D)

$f(x) = x^2 - kx + (b+c - \sqrt{ac} - \sqrt{bd}) = 0, \qquad a,b,c,d \to HP$

$\sqrt{ac} > b \quad \ldots\ldots(i)$

$\sqrt{bd} > c \quad \ldots\ldots(ii)$

$\therefore \sqrt{ac} + \sqrt{bd} > b + c$

$b + c - \sqrt{ac} - \sqrt{bd} < 0$

$\therefore \ f(0) = b + c - \sqrt{ac} - \sqrt{bd} < 0$

Hence, roots are of opposite sign.

Q.4 (C)

$x - 2\sqrt{x} + 1 - \sqrt{x}\,(a^2 - 3a) + a^2 - 3a + 4 = 0$

$(\sqrt{x}-1)^2 - (a^2 - 3a)(\sqrt{x}-1) + 4 = 0$

$(\sqrt{x}-1)^2 + 4 = (a^2 - 3a)(\sqrt{x}-1)$

$$\sqrt{x} - 1 + \frac{4}{\sqrt{x}-1} = a^2 - 3a$$

$$\sqrt{x}-1+\frac{4}{\sqrt{x}-1}\ \geq\ 4 \qquad \left\{\sqrt{x}-1>0\right\}$$

$\therefore\ a^2-3a\geq 4\ \Rightarrow\ a^2-3a-4\geq 0\ \Rightarrow\ (a-4)\,(a+1)\geq 0$

$\Rightarrow a\in(-\infty,-1]\cup[4,\infty)$ **Ans.**

Q.5 (A)

$$\sum_{n=1}^{\infty}\left(\frac{21}{(S_nS_{n+2}+S_{n-1}\cdot S_{n+1})-(S_nS_{n+1}+S_{n-1}S_{n+2})}\right)$$

$$=\sum_{n=1}^{\infty}\left(\frac{21}{S_n(S_{n+2}-S_{n+1})-S_{n-1}(S_{n+2}-S_{n+1})}\right)=\sum_{n=1}^{\infty}\left(\frac{21}{(S_{n+2}-S_{n+1})(S_n-S_{n-1})}\right)=\sum_{n=1}^{\infty}\left(\frac{21}{T_{n+2}\cdot T_n}\right)$$

$\therefore\ T_n=S_n-S_{n-1}=3n^2-2n-\left(3(n-1)^2-2(n-1)\right)=6n-5$

$T_{n+2}=6n+7$

Now, $$\sum_{n=1}^{\infty}\frac{21}{T_{n+2}\cdot T_n}=\sum_{n=1}^{\infty}\frac{21}{(6n+7)(6n-5)}=\frac{21}{12}\sum_{n=1}^{\infty}\left(\frac{1}{6n-5}-\frac{1}{6n+7}\right)$$

$$=\frac{21}{12}\left(\left(\frac{1}{1}-\frac{1}{13}\right)+\left(\frac{1}{7}-\frac{1}{19}\right)+\left(\frac{1}{13}-\frac{1}{25}\right)+\left(\frac{1}{19}-\frac{1}{31}\right)+\ldots\ldots\right)=\frac{21}{12}\times\frac{8}{7}=2$$ **Ans.**

Q.6 (B)

$$f(x)=\ln\sqrt{\sin^{-1}\left(\frac{x-1}{8}\right)}$$

$$0<\sin^{-1}\left(\frac{x-1}{8}\right)\leq\frac{\pi}{2}\ \Rightarrow\ 0<\frac{x-1}{8}\leq 1\ \Rightarrow\ 0<x-1\leq 8\ \Rightarrow\ 1<x\leq 9$$

$$g(x)=\frac{5x^2-10x+8}{x-2}=5x+\frac{8}{x-2}$$

g(x) is an integer when $x=3,4,6$

$\therefore\ \alpha=6$ and $\beta=3$

Required sum $=\dfrac{6}{1-\frac{1}{3}}=\dfrac{6}{\frac{2}{3}}=9$ **Ans.**

Q.7 (C)

$369=\frac{9}{2}(2\times 1+8\times d)$

$41=1+4d\Rightarrow d=10$

$ar^8=a+8d=1+8\times 10$

$r^8=81\ \Rightarrow r=\sqrt{3}$

$ar^6=1\,.\left(\sqrt{3}\right)^6=3^3=27$

$\therefore\ \frac{T_7}{3}=\frac{27}{3}=9$ **Ans.**

Q.8 (C)
Q.9 (A)
Q.10 (A)

$$\frac{[x]^2}{4} = x \cdot \{x\}$$

$$[x]^2 = 4(\{x\}+[x])\ \{x\} \Rightarrow [x]^2 = 4\{x\}\ [x] + 4\{x\}^2$$

$$4\{x\}^2 + 4\{x\}\ [x] - [x]^2 = 0$$

$$\left(\frac{2\{x\}}{[x]}\right)^2 + 2\left(\frac{2\{x\}}{[x]}\right) - 1 = 0 \qquad \frac{2\{x\}}{[x]} = r$$

$$r^2 + 2r - 1 = 0 \quad \Rightarrow \quad r = \frac{-2 \pm \sqrt{4+4}}{2} = \frac{-2 \pm 2\sqrt{2}}{2} \quad -1 \pm \sqrt{2}$$

$$\therefore\ r = \sqrt{2} - 1 \qquad \{\because x > 0 \Rightarrow r > 0\}$$

$$\frac{2\{x\}}{[x]} = \sqrt{2} - 1 \quad \Rightarrow \quad \{x\} = \left(\frac{\sqrt{2}-1}{2}\right)[x]$$

$$0 < \left(\frac{\sqrt{2}-1}{2}\right)[x] < 1 \quad \Rightarrow \quad 0 < [x] < \frac{2}{\sqrt{2}-1} \quad \Rightarrow \quad 0 < [x] < \underbrace{2\left(\sqrt{2}+1\right)}_{4.14}$$

$\therefore$ Possible values of $[x]$ are 1, 2, 3 and 4.

(ii) $1 < x < 2 \Rightarrow [x] = 1$

$$\{x\} = \frac{\sqrt{2}-1}{2} \Rightarrow x = \frac{\sqrt{2}-1}{2} + 1 = \frac{\sqrt{2}+1}{2}$$

$$\therefore \text{ Required product is, } \frac{\sqrt{2}+1}{2} \cdot \frac{1}{2} \cdot \frac{\sqrt{2}-1}{2} = \frac{1}{8}$$

(iii) $2 < x < 3 \Rightarrow [x] = 2$

$$\{x\} = \sqrt{2} - 1 \Rightarrow x = \sqrt{2} - 1 + 2 = \sqrt{2} + 1$$

$\therefore$ G.P. is , $\sqrt{2}+1,\ 1,\ \sqrt{2}-1, \ldots\ldots$

$$T_{20} = \left(\sqrt{2}+1\right)\cdot\left(\sqrt{2}-1\right)^{19} = \left(\sqrt{2}-1\right)^{18} = \tan^{18}\left(\frac{\pi}{8}\right) \qquad \textbf{Ans.}$$

Q.11 (ACD)

In A.P. and a H.P. having equal number of terms and their first are equal and their last terms are also equal, product of the terms equidistant from the begining of a sequence and from the end of another sequence is constant and which is equal to product of first and last terms of any sequence.

$\therefore$ (A) $a_4 h_8 = 36$

(C) $a_6 h_6 = 36 \Rightarrow \sqrt{a_6 h_6} = 6$

(D) $\dfrac{a_6 + h_6}{2} > \sqrt{a_6 h_6} \Rightarrow a_6 + h_6 > 12$ **Ans.**

Q.12 (ABD)

$x, y, z > 0$

$\therefore$ Applying A.M. $\geq$ G.M.

(A) $\dfrac{x+y+z}{3} \geq (xyz)^{1/3} \quad \Rightarrow \quad 3 \geq (xyz)^{1/3} \quad \Rightarrow \quad xyz \leq 27$

(B) $\dfrac{x+\frac{y}{2}+\frac{y}{2}+z}{4} \geq \left(x\cdot\dfrac{y^2}{4}\cdot z\right)^{1/4} \quad \Rightarrow \quad \dfrac{9}{4} \leq \left(\dfrac{xy^2z}{4}\right)^{1/4} \quad \Rightarrow \quad \dfrac{xy^2z}{4} \leq \left(\dfrac{9}{4}\right)^4 \quad \Rightarrow \quad xy^2z \leq 3^8 \cdot 2^{-6}$

(C) $\dfrac{\frac{x}{2}+\frac{x}{2}+y+\frac{z}{3}+\frac{z}{3}+\frac{z}{3}}{6} \geq \left(\dfrac{x^2}{4}\cdot y\cdot\dfrac{z^3}{27}\right)^{1/6}$

$x^2yz^3 \leq \left(\dfrac{3}{2}\right)^6 \times 4 \times 27 \Rightarrow \quad x^2yz^3 \leq 3^9 \cdot 2^{-4}$

(D) $\dfrac{\frac{x}{2}+\frac{x}{2}+\frac{y}{3}+\frac{y}{3}+\frac{y}{3}+\frac{z}{4}+\frac{z}{4}+\frac{z}{4}+\frac{z}{4}}{9} \geq \left(\dfrac{x^2}{4}\cdot\dfrac{y^3}{27}\cdot\dfrac{z^4}{256}\right)^{1/9}$

$x^2y^3z^4 \leq 4 \times 27 \times 256 \quad \Rightarrow \quad x^2y^3z^4 \leq 2^{10} \cdot 3^3$ **Ans.**

Q.13 (ABCD)

$x, y, 6$ are in A.P.

$\therefore \ y = \dfrac{x+6}{2}$

$6, z, 9x$ are in H.P.

$\therefore \ z = \dfrac{2\cdot 6\cdot 9x}{6+9x}$

$yz = 36 \quad \Rightarrow \quad \dfrac{x+6}{2}\cdot\dfrac{2\cdot 6\cdot 9x}{6+9x} = 36 \quad \Rightarrow \quad \dfrac{x(x+6)}{(2+3x)} = 2$

$\Rightarrow \quad x^2 + 6x = 4 + 6x \quad \Rightarrow \quad x = 2$

$y = 4, \quad z = 9$

$\therefore \quad zyx = 942 = 2 \times 3 \times 157$ **Ans.**

Q.14 (BCD)

$M = \underbrace{999\ldots\ldots 9}_{9\text{ times}} = 10^9 - 1$

$N = \underbrace{999\ldots\ldots 9}_{6\text{ times}} = 10^6 - 1$

$P = 999 = 10^3 - 1$

(A) $MP = (10^9 - 1)(10^3 - 1) = 10^{12} - 10^9 - 10^3 + 1 \neq N^2$

(B) $\dfrac{N}{P} = \dfrac{10^6 - 1}{10^3 - 1} = 10^3 + 1 = 1001$

(C) $N - 2P = 10^6 - 1 - 2(10^3 - 1) = 10^6 - 2 \cdot 10^3 + 1 = (10^3 - 1)^2 = P^2$

(D) $N - P^2 = 2P =$ even integer

Q.15 [0065]

$$a_{102} = a_1 + (101)d \qquad \Rightarrow \quad d = \frac{a_{102} - a_1}{101}$$

$$a_1^2 - a_2^2 + a_3^2 - a_4^2 + \ldots + a_{101}^2 - a_{102}^2$$

$$\Rightarrow -d\,(a_1 + a_2 + a_3 + \ldots\ldots + a_{101} + a_{102})$$

$$\Rightarrow -\left(\frac{a_{102} - a_1}{101}\right)\cdot\frac{102}{2}\left(\frac{a_1 + a_{102}}{2}\right)$$

$$= \frac{-102}{101}\cdot\frac{1}{4}\cdot 404 \times 406 = -17 \times 3 \times 2 \times 2 \times 7 \times 29 = -2^3 \cdot 3^1 \cdot 7^1 \cdot 17^1 \cdot 29^1$$

$\therefore$ p_is are 2, 3, 7, 17, 29 and b_i s are 3, 1, 1, 1, 1

$\Rightarrow \Sigma p_i + \Sigma b_i = (2 + 3 + 7 + 17 + 29) + (2 + 1 + 1 + 1 + 1) = 58 + 7 = 65$ **Ans.**

Q.16 [0023]

$$\frac{2}{b} = \frac{1}{a} + \frac{1}{c} \quad \Rightarrow \quad \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$$

$$\therefore \ \frac{3}{b} = 1 \quad \Rightarrow \quad b = 3$$

$$\frac{1}{a} + \frac{1}{c} = \frac{2}{3} \quad \Rightarrow \quad \frac{1}{a} > \frac{1}{3} \text{ and } \frac{1}{c} < \frac{1}{3} \quad \Rightarrow \quad a < 3 \text{ and } c > 3$$

If $a = 2$ is taken, then $\frac{1}{c} = \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \quad \Rightarrow \quad c = 6$

$\therefore$ number are 2, 3, 6, 12

If $a = 1$ is taken, then $\frac{1}{c} = \frac{2}{3} - 1 = -\frac{1}{3}$ Not possible.

$\therefore$ a, b, c and d are 2,3,6 and 12 respectively

$a + b + c + d = 23$ **Ans.**

Q.17 [0075]

$$T_r = -\frac{1}{2} + (r-1)\,1 = r - \frac{3}{2}$$

$$T_{r+1} = r - \frac{1}{2}, \quad T_{r+2} = r + \frac{1}{2}, \quad T_{r+3} = r + \frac{3}{2}$$

$$\sqrt{1 + T_r.T_{r+1}.T_{r+2}.T_{r+3}} = \sqrt{1 + \left(r - \frac{3}{2}\right)\left(r - \frac{1}{2}\right)\left(r + \frac{1}{2}\right)\left(r + \frac{3}{2}\right)} = \sqrt{1 + \left(r^2 - \frac{9}{4}\right)\left(r^2 - \frac{1}{4}\right)}$$

$$= \sqrt{r^4 - \frac{5}{2}r^2 + \frac{25}{16}} = \sqrt{\left(r^2 - \frac{5}{4}\right)^2} = r^2 - \frac{5}{4}$$

$$\left\{\sqrt{1 + T_r.T_{r+1}.T_{r+3}}\right\} = \left\{r^2 - \frac{5}{4}\right\} = \frac{3}{4}$$

$$\therefore \quad \sum_{r=2}^{101} \frac{3}{4} = \frac{3}{4} \times 100 = 75 \textbf{ Ans.}$$